# Lecture 9

- Orthogonal Bases
- The Gram-Schmidt process
- Orthogonal Matrices
- The QR factorization

Let $V$ be an inner product space.

Recall that $v, u \in V$ are called <u>orthogonal</u> if $\langle v, u \rangle = 0$

If $V = \mathbb{R}^n$ is the Euclidean space, then $\langle u, v \rangle = 0 \Leftrightarrow u \perp v$

Orthogonality is thus a generalization of the geometric property of perpendicularity. Bases consisting of mutually orthogonal vectors play a very important role in many applications because computations

1) become dramatically simpler
2) less prone to numerial instabilities

**Def** A basis $u_1 \dots u_n$ of $V$ is called orthogonal if $\langle u_i, u_j \rangle = 0 \quad \forall i \neq j$

An orthogonal basis $u_1 \dots u_n$ is called orthonormal if $\|u_i\| = 1 \quad \forall i$

<u>Example</u>: For $\mathbb{R}^n$ with dot product, the standard basis is orthonormal.

It is straightforward to convert any orthogonal basis into orthonormal:

$$u_i \rightsquigarrow \tilde{u}_i = \frac{u_i}{\|u_i\|}$$

normalized vector ($u_i \neq 0$, since $u_1 \dots u_n$ is basis)

A useful observation:

**Lemma 1** If $v_1 \dots v_k \in V$ are nonzero and mutually orthogonal

1. $v_i \neq 0$
2. $\langle v_i, v_j \rangle = 0 \quad i \neq j$

$\Rightarrow v_1 \dots v_k$ are lin. indep.

<u>Proof</u>: Suppose $v_1 \dots v_k$ are lin. dep. $\Rightarrow \exists$ nontrivial lin combination

$$\alpha_1 v_1 + \dots + \alpha_k v_k = 0 \quad \Rightarrow \quad 0 = \langle \alpha_1 v_1 + \dots + \alpha_n v_k, v_i \rangle = \alpha_i \underbrace{\|v_i\|^2}_{\neq 0} \Rightarrow \alpha_i = 0 \quad \square$$

Therefore, any collection of nonzero orthogonal vectors forms a basis for its span.

**Theorem 1** Let $v_1 \dots v_k \in V$, $v_i \neq 0$ and $\langle v_i, v_j \rangle = 0$.

Then $v_1 \dots v_k$ form an orthogonal basis for $W = \text{span}\{v_1 \dots v_k\} \subset V$, $\dim W = k$

Moreover, if $\dim V = k \Rightarrow v_1 \dots v_k$ form an orthogonal basis for $V$.

<u>Question</u>: What is the advantage of an orthogonal (orthonormal) basis? (over other bases)

If $v_1 \dots v_n$ is a basis of $V$, then any vector of $V$ can be expressed as a lin. combination: $v = \alpha_1 v_1 + \dots + \alpha_n v_n$. Coefficients $\{\alpha_i\}$ are called the coordinates of $v$ in basis $\{v_1 \dots v_n\}$. In general, finding coordinates is a difficult task: we need to solve a system of lin. equations:

$$\left[ v_1 \middle| v_2 \middle| \dots \middle| v_n \right] \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix} = \left[ v \right]$$

In applications, $n$ may be large $\Rightarrow$ computing the solutions may require a considerable amount of time and effort.

But, if $\{v_1 \dots v_n\}$ is orthogonal/orthonormal $\Rightarrow$ it is easy to find the coord. in this basis.

**Theorem 2** Let $u_1 \dots u_n$ be an orthonormal basis of $V$.
Then the coordinates of any $v \in V$, $v = \alpha_1 u_1 + \dots + \alpha_n u_n$,
are given by $\alpha_i = \langle v, u_i \rangle$, $i = 1 \dots n$.
Moreover, $\|v\| = \sqrt{\sum \alpha_i^2} = \sqrt{\sum_{i=1}^n \langle v, u_i \rangle^2}$

**Proof** Let's simply check that $\langle v, u_i \rangle = \alpha_i$

$$\langle v, u_i \rangle = \left\langle \sum_{j=1}^n \alpha_j u_j, u_i \right\rangle = \sum_{j=1}^n \alpha_j \underbrace{\langle u_j, u_i \rangle}_{= \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}} = \alpha_i$$

Similar computation for the norm:

$$\|v\|^2 = \langle v, v \rangle = \left\langle \sum_i \alpha_i u_i, \sum_j \alpha_j u_j \right\rangle = \sum_{i,j} \alpha_i \alpha_j \langle u_i, u_j \rangle = \sum_{i=1}^n \alpha_i^2 \qquad \square$$

**Remark** If $u_1 \dots u_n$ is orthogonal (but not necessaril orthonormal)

$$\Rightarrow \alpha_i = \frac{\langle v, u_i \rangle}{\|u_i\|^2} \quad \text{and} \quad \|v\| = \sqrt{\sum_{i=1}^n \left( \frac{\langle v, u_i \rangle}{\|u_i\|} \right)^2}$$

**Example** One of the most important examples of an orthogonal basis is provided by the basic trigonometric functions.

Let $T^{(n)} = \left\{ T(x) = \sum_{0 \le j+k \le n} a_{jk} (\sin x)^j (\cos x)^k, \ x \in [-\pi, \pi], \ a_{jk} \in \mathbb{R} \right\}$, with $L^2$ inner product

trigonometric polynomials of deg $\le n$

What is $\dim T^{(n)}$?
What is a basis in $T^{(n)}$?

The individual monomials span $T^{(n)}$ : $T^{(n)} = \operatorname{span}\left\{ (\sin x)^j (\cos x)^k : 0 \le j+k \le n \right\}$

Test question: do they form a basis? No, since $(\sin x)^2 + (\cos x)^2 = 1$.

It can be shown that $\underbrace{1, \cos x, \sin x, \cos 2x, \sin 2x, \dots, \cos nx, \sin nx}_{(2n+1) \text{ functions}}$ also span $T^{(n)}$. (Problem 3.6.21)

Indeed: $T(x) = \sum_{0 \le j+k \le n} a_{jk} \left( \frac{e^{ix} - e^{-ix}}{2i} \right)^j \left( \frac{e^{ix} + e^{-ix}}{2} \right)^k = \sum_{\substack{0 \le j+k \le n \\ m=-n}}^{n} b_{jkm} e^{imx} =$

$$= \sum_{\substack{0 \le j+k \le m \\ m=-n}}^{n} b_{jkm} (\cos mx + i \sin mx) = \sum_{m=-n}^{n} c_m \cos(mx) + \sum_{m=-n}^{n} d_m \sin(mx).$$

Moreover, these functions form an orthogonal basis of $T^{(n)}$ (with respect to $L^2$ norm on $[-\pi, \pi]$)

To prove orthogonality is straighforward: just compute $\langle f, g \rangle = \int_{-\pi}^{\pi} f(x) g(x) \, dx$.

The fact that the functions form a basis follows from Th 1.

As a consequence, $\dim T^{(n)} = 2n+1$.

# The Gram-Schmidt Process

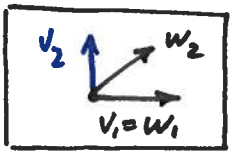
After we become convinced that orthogonal/orthonormal bases are useful, a natural question arises: How to construct them for arbitrary inner product space? A practical algorithm was first discovered by P.-S. Laplace (and then rediscovered by Gram & Schimdt)

Let $V$ be a finite-dimensional inner product space, and let $w_1 \dots w_n$ be some basis of $V$, $\dim V = n$. <u>Goal</u>: construct an orthogonal basis $v_1 \dots v_n$ from $w_1 \dots w_n$

<u>Algorithm</u>: (basic algorithm)

① $v_1 = w_1$ (Note that $v_1 \neq 0$, since $w_1$ is an element of a basis)

Let us look for $v_2$ in span $(v_1 = w_1, w_2)$. Since $v_2 \nparallel v_1 = w_1$, we can look for $v_2$ in the following form: $v_2 = w_2 + \alpha v_1$.




span $(w_1, w_2)$

Using condition $\langle v_1 v_2 \rangle = 0$, we have: $\langle w_2 + \alpha v_1, v_1 \rangle = 0$

$\Rightarrow \langle w_2, v_1 \rangle + \alpha \|v_1\|^2 = 0 \Rightarrow \alpha = -\dfrac{\langle w_2, v_1 \rangle}{\|v_1\|^2}$.

② $v_2 = w_2 - \dfrac{\langle w_2, v_1 \rangle}{\|v_1\|^2} v_1$ (Note that $v_2 \neq 0$, since $w_1 = v_1$ and $w_2$ are lin. indep.)

Similarly, we can look for $v_3$ in the form $v_3 = w_3 + \alpha_1 v_1 + \alpha_2 v_2$

We need to find $\alpha_1$ and $\alpha_2$ such that $\langle v_3, v_1 \rangle = 0$ and $\langle v_3, v_2 \rangle = 0$.

$\Rightarrow$ ③ $v_3 = w_3 - \dfrac{\langle w_3, v_1 \rangle}{\|v_1\|^2} v_1 - \dfrac{\langle w_3 v_2 \rangle}{\|v_2\|^2} v_2$

| | |
|---|---|
| $\langle w_3, v_1 \rangle + \alpha_1 \|v_1\|^2 = 0$ | $\langle w_3, v_2 \rangle + \alpha_2 \|v_2\|^2 = 0$ |
| $\alpha_1 = -\dfrac{\langle w_3, v_1 \rangle}{\|v_1\|^2}$ | $\alpha_2 = -\dfrac{\langle w_3, v_2 \rangle}{\|v_2\|^2}$ |

Again $v_3 \neq 0$, since $v_3$ is a nontrivial linear combination of $w_1\ w_2\ w_3$.

Suppose now, that we constructed the mutually orthogonal vectors $v_1 \dots v_{k-1} \in$ span $(w_1 \dots w_{k-1})$. The next orthogonal basis element $v_k$ can be constructed in the following form: $v_k = w_k + \sum_{i=1}^{k-1} \alpha_i v_i$. We need to find $\alpha_1 \dots \alpha_{k-1}$ from $(k-1)$ orthogonality conditions: $\langle v_k, v_j \rangle \overset{!}{=} 0$, $j = 1 \dots k-1$. $\Rightarrow \langle w_k, v_j \rangle + \alpha_j \|v_j\|^2 = 0$

$\Rightarrow \alpha_j = -\dfrac{\langle w_k, v_j \rangle}{\|v_j\|^2}$

④ $$\boxed{v_k = w_k - \sum_{i=1}^{k-1} \frac{\langle w_k, v_i \rangle}{\|v_i\|^2} v_i}$$ (*)

Gram-Schmidt formula

Thus, the Gram-Schmidt process (*) defines an explicit, recursive procedure for constructing the orthogonal basis $v_1 \dots v_n$.

If we need an orthonormal basis $\Rightarrow$ we simply rescale $u_i = \dfrac{v_i}{\|v_i\|}$, $i = \overline{1,n}$

Recall that any finite-dim. vector space (except $\{0\}$) admits a basis.
If $V$ is endowed with an inner product $\overset{GS}{\Longrightarrow}$ it has an orthogonal basis

<u>Theorem 3</u> Any non-zero finite-dim inner product space has an orthonormal basis.

The basic Gram-Schmidt algorithm is intuitive, but it has a serious drawback: in practical large scale computations it is subject to numerical instabilities. Accumulating round-off errors may lead to inaccuracies, non-orthogonal vectors. Fortunately, there is a simple rearrangement of the calculations that is numerically robust and most often used in practice.

<u>Numerically stable Gram-Schmidt algorithm</u>: $(w_1 \dots w_n) \longrightarrow \overbrace{(u_1 \dots u_n)}^{\text{orthonormal}}$

① $u_1 = \dfrac{w_1^{(1)}}{\|w_1^{(1)}\|}$ (as before) $\qquad W = \{w_1^{(1)}, \dots, w_n^{(1)}\}$

Before computing $u_2$, we modify $W = \{w_2^{(1)} \dots w_n^{(1)}\}$ as follows:

$$w_k^{(2)} = w_k^{(1)} - \langle w_k^{(1)}, u_1 \rangle u_1 \qquad k = 2, \dots, n$$

(in the basic alg. we did this only for $k=2$ and $w_k^{(2)}$ was $v_2$)

(so that $w_2^{(2)} \dots w_n^{(2)}$ are orthogonal to $u_1$:
$\langle w_k^{(2)}, u_1 \rangle = \langle w_k^{(1)}, u_1 \rangle - \dfrac{\langle w_k^{(1)}, u_1 \rangle \|u_1\|^2}{\text{[redacted]}} = 0$)

=

② $u_2 = w_2^{(2)} / \|w_2^{(2)}\|$

Before computing $u_3$, we modify $\{w_3^{(2)}, \dots, w_3^{(2)}\}$ as follows:

$$w_k^{(3)} = w_k^{(2)} - \langle w_k^{(2)}, u_2 \rangle u_2$$

(Again so that $w_3^{(3)} \dots w_n^{(3)}$ are $\perp u_2$)

③ $u_3 = w_3^{(3)} / \|w_3^{(3)}\|$

And so on. $\Longrightarrow$

$$\boxed{\begin{aligned} u_j &= \frac{w_j^{(j)}}{\|w_j^{(j)}\|} \qquad & w_j^{(1)} &= w_j \\ & & j &= 1, \dots, n \\ & & k &= j+1, \dots, n \\ w_k^{(j+1)} &= w_k^{(j)} - \langle w_k^{(j)}, u_j \rangle u_j & & \end{aligned}}$$

<u>ACM 106 more details</u>

This results into a numerically stable computation of the same orthonormal basis vectors $u_1 \dots u_n$

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<u>Orthogonal Matrices</u> We discussed orthogonal vectors. There are also orthogonal matrices.

~~Remark (after def.) A matrix $Q$ is called orthogonal if its columns form an orthonormal basis of $\mathbb{R}^n$.~~

More formally: <u>Def</u> A square matrix $Q$ is called orthogonal if $Q^T Q = I$.

Orthogonal matrices appear in a wide range of applications: geometry, physics, PDEs.

In particular :

- rotational motions o bodies in 3D are described by orthogonal matrices
  $\Rightarrow$ used in rigid body mechanics, computer graphics and animation.
- orthogonal matrices are the key ingredient in the one of the most important methods of numerical linear algebra : the QR algorithm for computing eigenvalues.

Orthogonality condition $Q^TQ = I$ immediately implies, that it is very easy to find the inverse of an orthogonal matrix : $\boxed{Q^{-1} = Q^T}$

Here is a list of several properties of orthogonal matrices.

① $Q$ is orthogonal $\Longleftrightarrow$ its columns form an orthonormal basis of $\mathbb{R}^n$

Indeed, if $u_1 \dots u_n$ are the columns of $Q \Rightarrow Q^TQ = I$ means exactly $\langle u_i, u_j \rangle = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$

Remark "Orthonormale" matrix seems to be a better name, but traditionally they are called orthogonal matrices.

② An orthogonal matrix has determinant $\det Q = \pm 1$

Indeed, $1 = \det(Q^TQ) = \det Q^T \cdot \det Q = (\det Q)^2$.

Remark If $\det Q = 1$, it is called "special".

③ If $Q_1$ and $Q_2$ are orthogonal $\Rightarrow$ so is $Q_1Q_2 = Q$.

Indeed, $Q^TQ = (Q_1Q_2)^TQ_1Q_2 = Q_2^T \underbrace{Q_1^TQ_1}\, Q_2 = Q_2^TQ_2 = I$

Remark This property says that the set of all orthogonal matrices forms a group

The Gram-Schmidt procedure can be reinterpreted as a matrix factorization.

## The QR factorization

Let $w_1 \dots w_n$ be a basis of $\mathbb{R}^n$ and let $u_1 \dots u_n$ be the corresponding orthonormal basis that results from the Gram-Schmidt process.

The original $w_1 \dots w_n$ can be expressed in terms of $u_1 \dots u_n$ as follows :

$$\begin{cases} w_1 = r_{11}u_1 \\ w_2 = r_{12}u_1 + r_{22}u_2 \\ w_3 = r_{13}u_1 + r_{23}u_2 + r_{33}u_3 \\ \vdots \\ w_n = r_{1n}u_1 + \dots + r_{nn}u_n \end{cases}$$

Let $A = [w_1|w_2|\dots|w_n]$, $Q = [u_1|u_2|\dots|u_n]$

$\uparrow$ orthogonal

$\Longleftrightarrow \boxed{A = QR}$

nonsingular

$$R = \begin{bmatrix} r_{11} & r_{12} & \dots & r_{1n} \\ & r_{22} & \dots & r_{2n} \\ & & \ddots & \vdots \\ & & & r_{nn} \end{bmatrix} \leftarrow \text{upper triangular.}$$

**Theorem 4** (QR factorization)

Any nonsingular matrix $A$ can be factored $A = QR$ into the product of an orthogonal matrix $Q$ and an upper triangular matrix $R$.

**Remark** The factorization is unique if $r_{ii} > 0$ (exercise 5.3.32)
(can be seen from the basic algorithm)

**Remark** The QR factorization can be used as an alternative to Gaussian Elim. to solve linear systems:

$$Ax = b \Rightarrow QRx = b \Rightarrow Rx = Q^T b$$

Since $R$ is upper triangular, the linear system can be solved for $x$ by Back Substitution.

- This approach is more computationally expensive, but
- less prone to inaccuracies, more numerically stable

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